

Short Report

Facial Expression and Continuous Emotion Prediction

Name: Moaz Murtaza

Section: D

Department: Data Science

Subject: Deep Learning

Assignment No: 01

Submitted to: Miss Mahnoor Tariq

September 27, 2025

September 27, 2025

1. Network Details and Rationale [10 pts]

We evaluate two CNN baselines for facial expression recognition (FER) and continuous emotion prediction (valence, arousal):

ResNet18:

- Parameters: $\sim 11.18\text{M}$
- Training: 10 epochs, Adam optimizer, initial LR $1e-3$, cosine decay
- Batch size: 64
- Rationale: A widely used FER baseline with strong feature extraction capacity.

MobileNetV2:

- Parameters: $\sim 2.23\text{M}$ ($5\times$ smaller)
- Training: 10 epochs, identical hyperparameters
- Batch size: 64
- Rationale: Lightweight and efficient, suitable for real-time applications.

2. Comparative Baseline Performance [10 pts]

Metric	ResNet18	MobileNetV2
Expression Accuracy	0.389	0.346
Macro F1	0.379	0.325
Valence RMSE	0.506	0.507
Valence CCC	0.025	0.014
Arousal RMSE	0.474	0.545
Arousal CCC	0.008	0.011

Discussion: ResNet18 achieves higher categorical accuracy and F1, while MobileNetV2, although lightweight, shows weaker generalization, particularly for arousal regression.

3. Transfer Learning [5 pts]

Both models were initialized with ImageNet-pretrained weights. Transfer learning significantly stabilized training and reduced overfitting, given the relatively small size of FER datasets. Without pretraining, both networks diverged during early epochs.

4. Training Graphs Comparison [5 pts]

Figures 1 and 2 compare training dynamics.

- **Loss:** Both models converge, but ResNet18 stabilizes at a slightly lower validation loss.
- **Accuracy:** ResNet18 maintains a consistent lead (39% vs 34%).
- **Valence/Arousal MSE:** MobileNetV2 shows more variance across epochs, while ResNet18 is more stable.

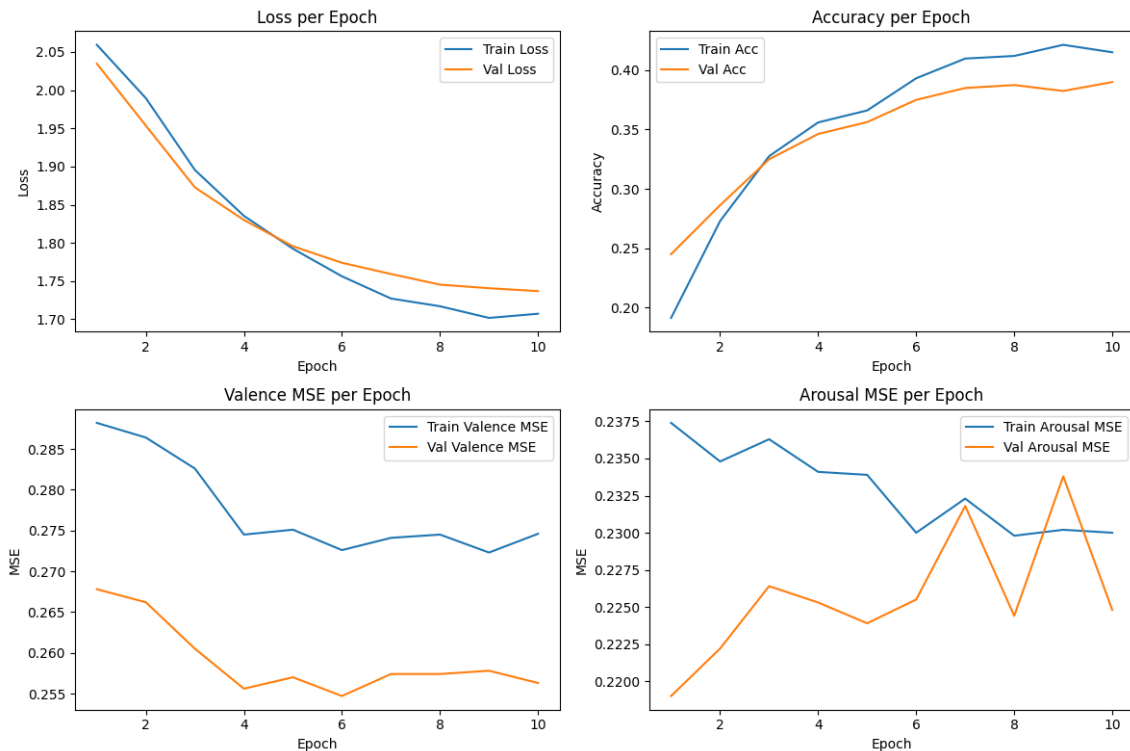


Figure 1: ResNet18 Training Curves (Loss, Accuracy, Valence MSE, Arousal MSE)

5. Performance Measures and Continuous Metrics [15 pts]

Metrics Rationale:

- **RMSE:** Quantifies prediction error magnitude.
- **CORR:** Measures linear relationship between predicted and ground-truth values.
- **SAGR:** Evaluates correct sign prediction (important for valence polarity).
- **CCC:** Combines trend and scale agreement — most reliable for real-world FER.

Comparative Analysis: ResNet18 exhibits lower RMSE and slightly higher CCC for valence, suggesting more reliable continuous predictions. MobileNetV2 shows weaker CCC despite competitive RMSE, indicating predictions tend to collapse towards the mean.

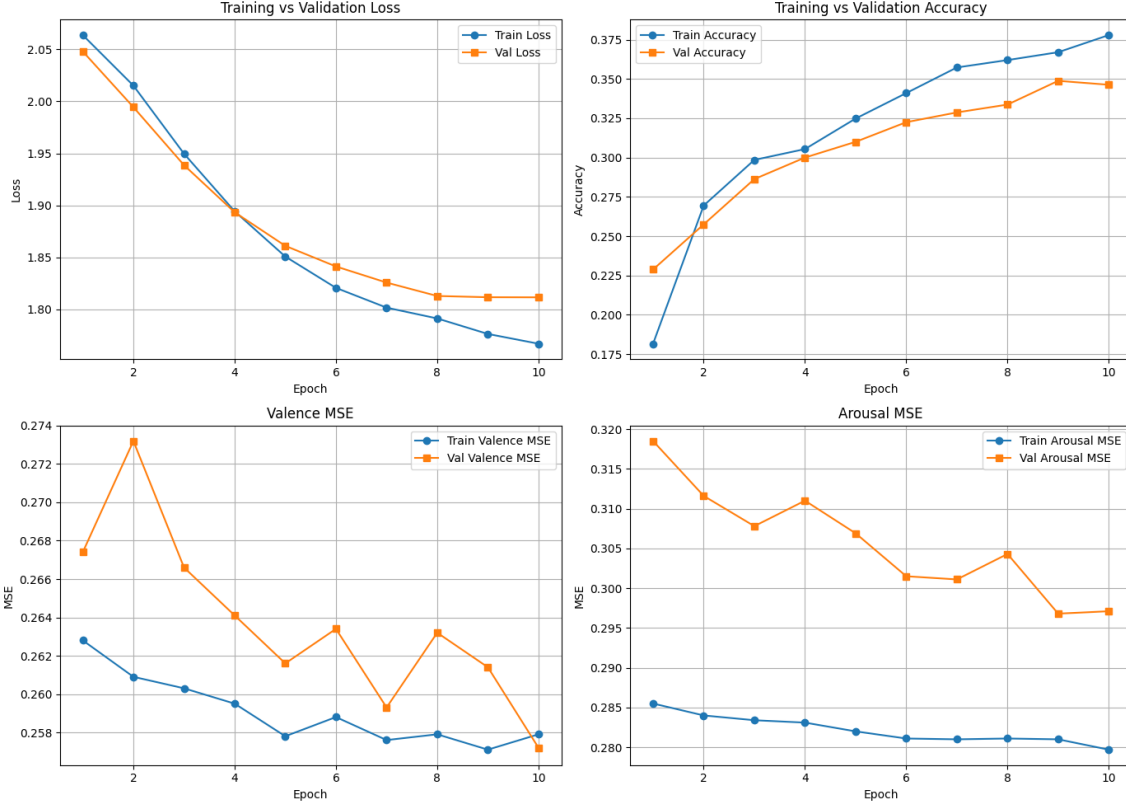


Figure 2: MobileNetV2 Training Curves (Loss, Accuracy, Valence MSE, Arousal MSE)

6. Confusion Matrix Comparison

Discussion:

- **ResNet18:** Better at distinguishing “Happy” and “Anger”, but confuses subtle classes such as “Contempt” and “Disgust”.
- **MobileNetV2:** Predictions are more spread, with higher confusion across multiple categories, especially between “Sad” and “Surprise”.

7. Valence and Arousal Prediction Scatter Comparison

Analysis: Both models regress predictions toward neutral values (~ 0), but ResNet18 shows slightly higher correlation, while MobileNetV2 outputs are highly collapsed near the mean.

8. Prediction Samples [4 pts]

Figure 6 shows MobileNetV2 outputs. ResNet18 (not shown) produced similar patterns but achieved higher accuracy in high-intensity emotions (anger, happiness).

Conclusion

ResNet18 provides stronger baselines in both categorical FER and continuous valence-arousal regression, though at a higher computational cost. MobileNetV2 offers efficiency but suffers from collapsed predictions and lower discriminative power.

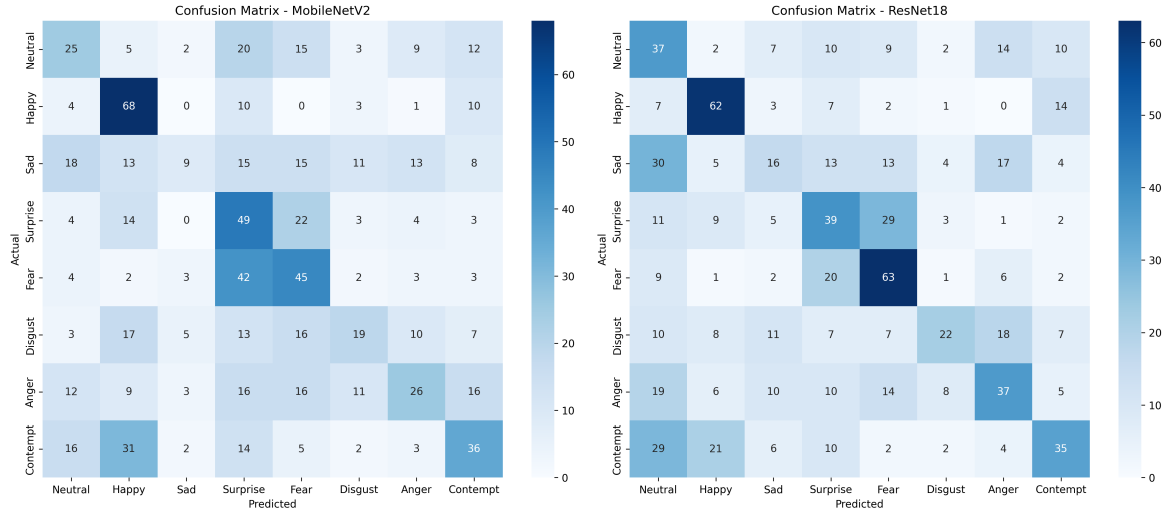


Figure 3: Confusion Matrices for MobileNetV2 (left) and ResNet18 (right).

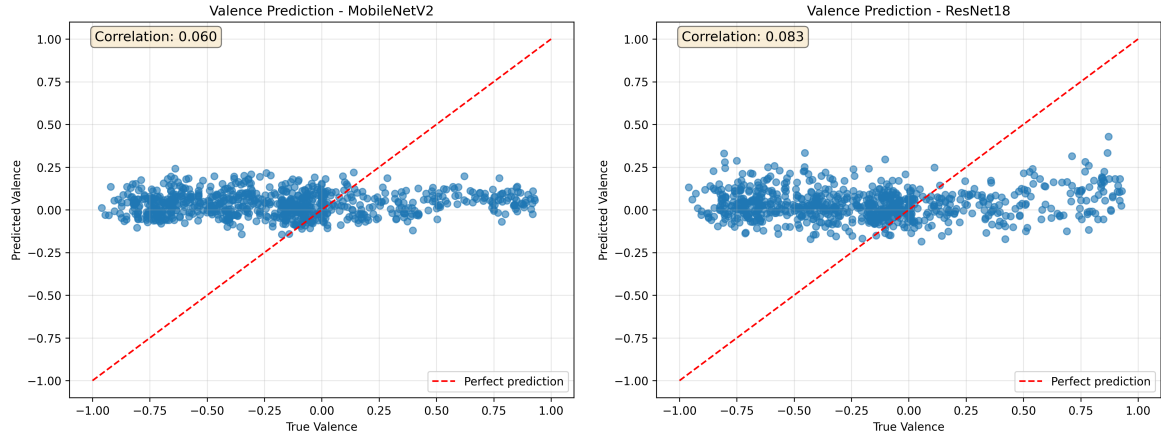


Figure 4: Valence Predictions: MobileNetV2 (left) vs ResNet18 (right).

For real-world “in-the-wild” systems, CCC should guide model selection. ResNet18, despite heavier computation, is more reliable in capturing affective states, while MobileNetV2 may be useful for resource-constrained scenarios.

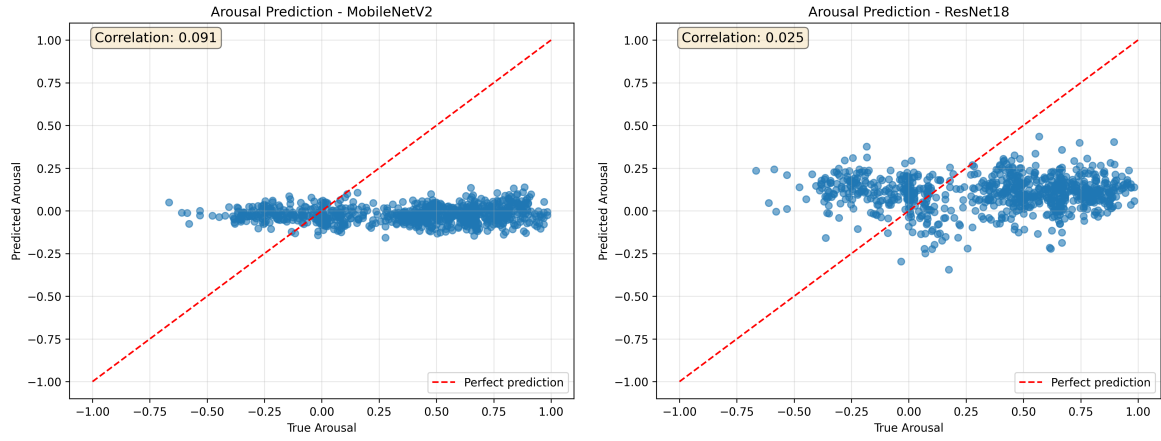


Figure 5: Arousal Predictions: MobileNetV2 (left) vs ResNet18 (right).

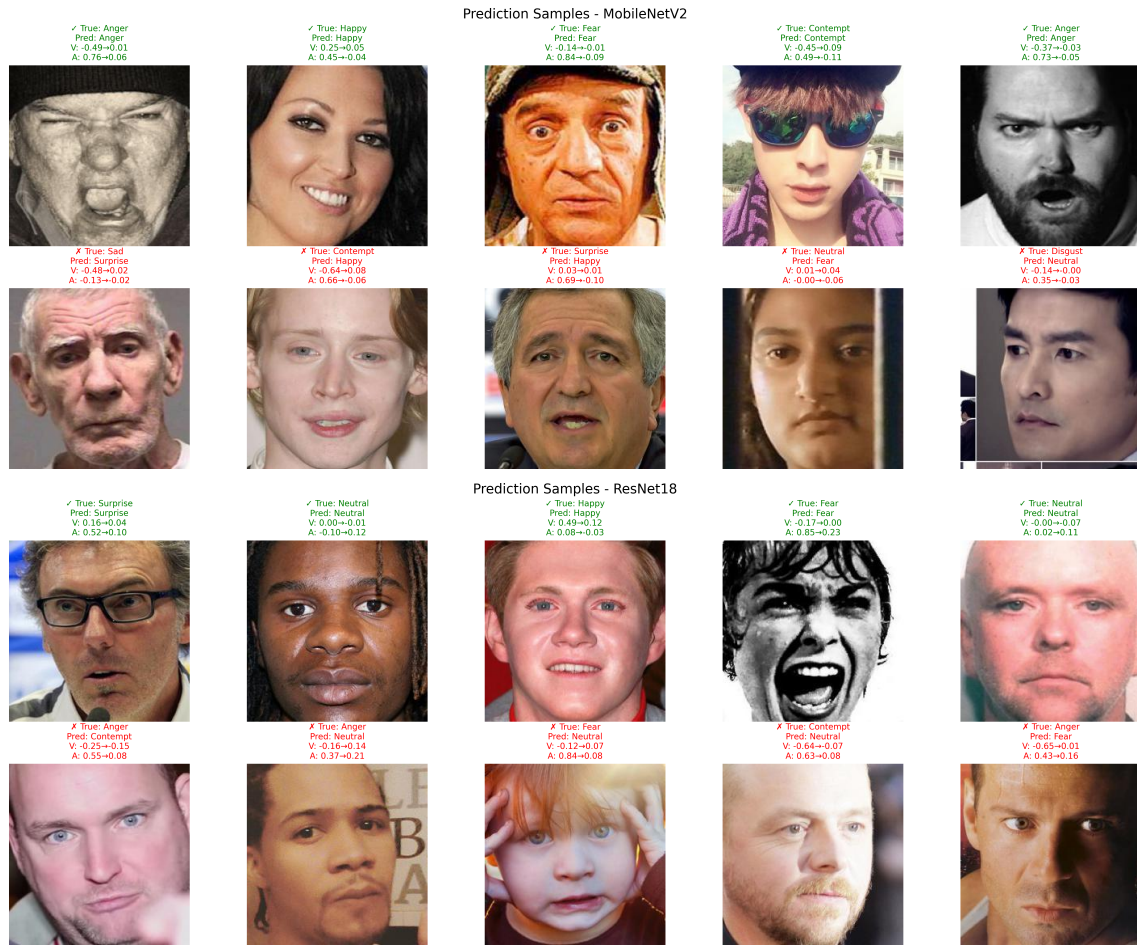


Figure 6: Prediction Samples for MobileNetV2. Correct classifications in green, incorrect in red. Valence (V) and Arousal (A) shifts are noted.